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§ 8.1

- Zipping through chp. 8 to get to limits

ordered field

Let $F (= \mathbb{Q}, \mathbb{R}, \mathbb{C})$ be an ordered field & suppose we have $<, >, \leq$ & $\geq$ behaving in ways we'd expect.

$\boxed{\text{Thm. 6}}$ (Trichotomy)

If $x, y \in F$ then exactly 1 of the following occurs...

i) $x = y$ ii) $y < x$ iii) $x < y$

§ 8.2

Absolute Value

$$|x| \overset{\text{def}}{=} \begin{cases} x & x \geq 0 \\ \end{cases}$$

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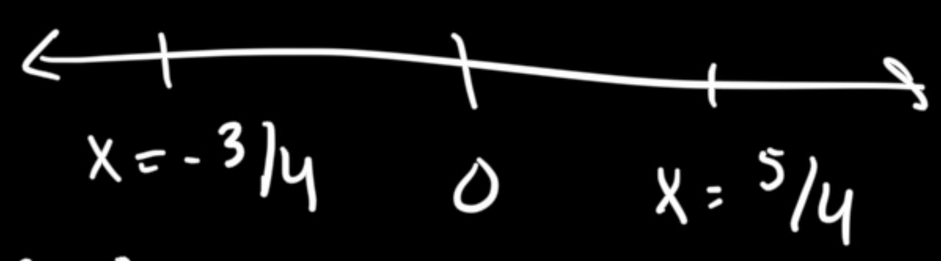
§8.2

Absolute Value

$|x|$ of $x \in F$ is defined to be $|x| \overset{\text{def}}{=} \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$



$|-3/4|$
$= -(-3/4)$
$= 3/4$

$|5/4| = 5/4$

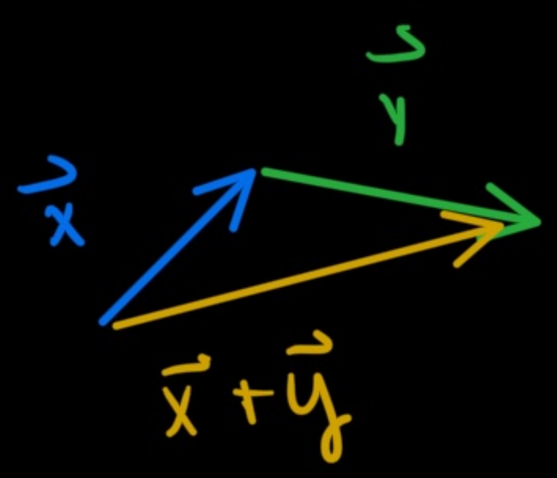
"The distance from x to 0"

$|x-y| \to$ "distance between x & y"

on hw 11

$\boxed{\text{Thm. 30}}$ (triangle inequality)

If $x, y \in F$ then $|x+y| \leq |x| + |y|$.



pf

If $x \geq 0$, $|x| = x$

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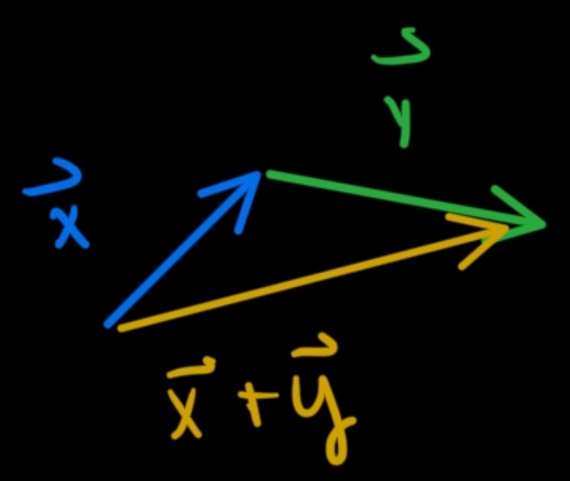
on hw 11

Thm. 30 (triangle inequality)

If $x, y \in F$ then $|x+y| \leq |x| + |y|$.



<u>pf</u>

If $x \geq 0$, $|x| = x$

If $x < 0$, $|x| > x$

This tells us $|x| \geq x$. Hence $x+y \leq |x|+y$
$\leq |x|+|y|$.

Likewise, $-x-y \leq |x|-y \leq |x|+|y|$.

If $x+y \geq 0$, then $|x+y| = x+y$.

$|x+y| = x+y \leq |x|+|y|$. If $x+y \leq 0$, then

$|x+y| = -x-y \leq |x|+|y|$. $\therefore$ $|x+y| \leq |x|+|y|$

$\square$

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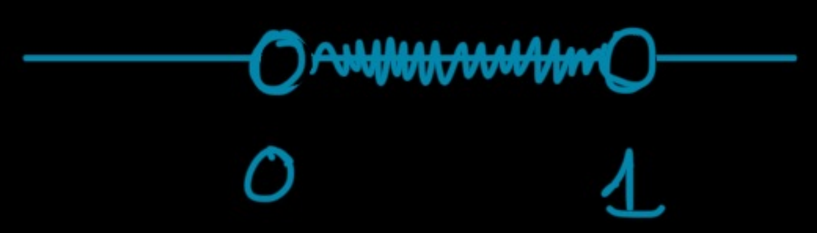
Def. 7 (Intervals)

For $x, y \in F$, with $x < y$ define

$$(x, y) \overset{\text{def}}{=} \{ z \in F : x < z < y \}.$$

For example $0, 1 \in \mathbb{R}$ is



$\boxed{\text{Thm. 32}}$ (Density)

"Between any 2 rational numbers $\exists$ another rational number"

$\Rightarrow$ Let $x, y \in F$, $x < y$. $\exists\, z \in F$ with $x < z < y$.

In other words $(x, y) \neq \emptyset$.

$$\overset{\shortparallel}{\{ z \in F : x < z < y \}}$$

ex

$$\left(\frac{314159}{100000}, \frac{22}{7} \right)$$

pf

Consider $z = \dfrac{x+y}{2} = 2^{-1}(x+y)$.

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pf

Consider $z = \frac{x+y}{2} = 2^{-1}(x+y)$.

Since $x < y$, $x+x < x+y < y+y$.

Thus $2x < x+y < 2y$. As $2 \neq 0$, $2^{-1} = \frac{1}{2} \in F$.

Since $2 > 0$, $\frac{1}{2} > 0$. Thus

$$\frac{1}{2}(2x) < \frac{1}{2}(x+y) < \frac{1}{2} \cdot 2y \Rightarrow x < \frac{x+y}{2} < y.$$

$\square$

§8.3 The Archimedian Property

Def. 10

F is Archimedian if $\mathbb{Q} \subset F$ as an ordered subfield & if $x, y \in F$, $x > 0$, then $\exists$ $n \in \mathbb{N} \subset F$: $nx \geq y$.

Choose $x = 1$ to see $\mathbb{N}$ are unbounded in F.

really big #

small #

take any $n \in \mathbb{N}$

$1/y > 1$

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Thm. 37

If F is an Archimedean ordered field, then $\mathbb{Q}$ is dense in F. In other words, if $x<y \in F$ $\exists$ $r \in \mathbb{Q}: x<r<y$.

almost like betweeness

§8.5 Infinite Sequences & Limits

(*)

<u>def</u> An infinite sequence with values in F is a function whose domain is $\mathbb{N}$ & co-domain is F.

We write a_i to denote the image of $i \in \mathbb{N}$ under this function.

ex

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$i \in \mathbb{N}$ under this function.

ex

a) $a_i = i$ is $\{0, 1, 2, 3, \ldots\}$ in $\mathbb{Q}$

b) $a_i = \frac{1}{i}$ with $i > 0$ $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$

$a_i = \frac{1}{i+1}$

c) $b_i = \frac{-1}{i^2}$ w/ $i > 0$ $\{-1, -\frac{1}{4}, -\frac{1}{9}, -\frac{1}{16}, \ldots\}$

d) $c_i = (-1)^i \frac{1}{i^2 - i}$ w/ $i > 1$ $\{\frac{1}{2}, -\frac{1}{6}, \frac{1}{12}, -\frac{1}{20}, \ldots\}$

$(-1)^3 \frac{1}{3^2 - 3}$

$-1 \frac{1}{6} = -\frac{1}{6}$